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Pioneering communities in winter

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Plate 2: Stranded ostracods at K1 in May. Photo: Patrick Armitage.

Pioneering communities in winter

The initial observations arose from simple curiosity: what animals colonised these new habitats during the winter period? Patrick Armitage and Gloria Tapia look at ephemeral ponds.

In the winter period, the fields in the Frome Valley adjacent to the East Stoke River Laboratory are subject to some degree of flooding. This can involve the whole floodplain but more often than not results in a series of small flooded pools. These pools may arise simply from heavy rainfall and are unconnected to the river, except in the most extreme floods, but others may receive water from the bankfull River Frome and adjacent Mill Stream.

A feature common to most of these habitats is ephemerality. The unconnected pools generally have a longer life than those receiving water from the main river and Mill Stream during high flow periods. The period of wetting is concentrated in the winter period although summer flooding can occur. The initial observations arose from simple curiosity as to what animals colonised these new habitats during the winter period? How long did the habitats last and what changes occurred over time?

The literature on temporary waters and the adaptations of animals to them is extensive (Williams, 2005) and floodplain wetlands have been the subject of a range of studies including improvement of habitat for wading birds, the effects of irrigation on terrestrial vegetation and macroinvertebrates and more detailed work on colonisation and persistence of macroinvertebrate communities (Ausden *et al.*, 2001, Woodcock *et al.*, 2005, Schirmel *et al.*, 2014, Tronstad *et al.*, 2005) but few have examined small ephemeral winter flooded pools. These are a particularly challenging area of study due to their relatively random occurrence, and short period of existence. This article describes observations of the macroinvertebrate faunal communities taken sporadically between 1999 and 2014

in the River Frome floodplain adjacent to the River Laboratory at East Stoke Dorset (Figure 1). Two sets of data examine temporal change in two separate sites (A and D) and the remaining observations are 'spot samples' from 32 pools not exceeding 30 cm in depth and 120 m² in area and include a sample from a flooded winter pool 'C', about 4 km upstream (not shown in Fig.1).

Sampling consisted of sweeping a pond net over a standard distance of approximately 4 m.

Temporal change

Site A, a flooded pool (circular, with a diameter of 12 m and a maximum depth of 30 cm),

situated 10 m from the River Frome, was examined on 25 December 1999, a week after wetting and on three other occasions before the habitat dried up on 12 January 2000 (Figure 2).

Initially there was little change and the fauna was dominated by terrestrial organisms (mainly slugs, arachnids and Collembola) but 16 days after first wetting, aquatic fauna appeared in the samples. These included large numbers of Chironomidae (*Limnophyes* and *Metriocnemus* sp) and the psychodid *Pericoma trivialis*. High densities of the caddis *Limnephilus auricula*, whose larvae overwinter, were also recorded together with the coleopteran larvae of *Agabus* sp. and adult *Helophorus grandis*. These changes in faunal abundance were accompanied by a massive increase in microcrustaceans (Copepoda and Ostracoda) but within two days the habitat was dry.

A second set of observations between November 2010 and March 2011 were carried out on a small pool D adjacent to Ds, which never received river water and with maximum dimensions of 25 m long, 4 m wide and 25 cm deep. The pool first filled on the 12th of November and the first samples were taken on the 19th (Figure 3).

The pool dried out at the end of November after freezing. Alternating snowfall, thaw and ice in December meant that there was no open water until the pool filled with melting snow and by 1 January it was full again. Levels continued to fluctuate wildly with consequent reductions in surface area (Plate 1).

Within seven days of filling, the fauna was still dominated with terrestrial species. Early in January indeterminate juvenile *Limnephilus* larvae were found and by mid January Coleoptera (*Helophorus grandis*, *H. brevipalpis*, *H. flavipes*, *Ochthebius dilatatus*)

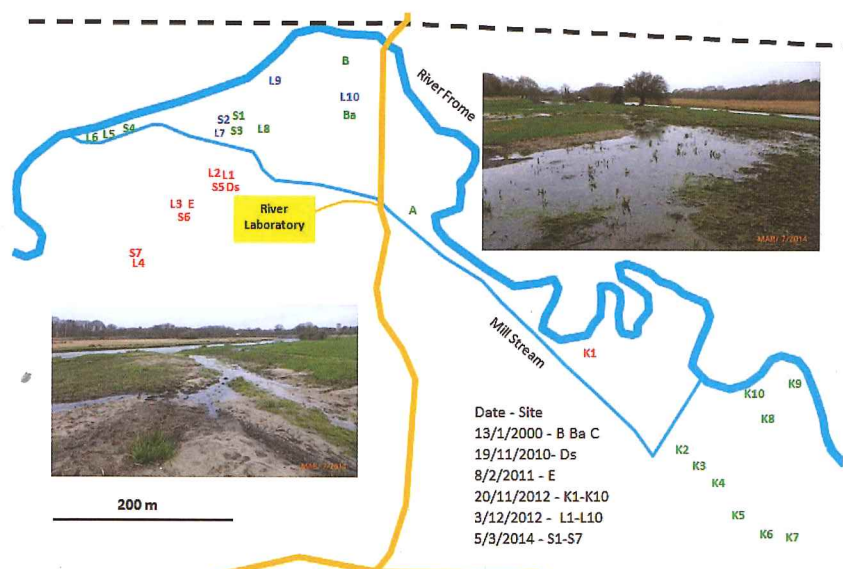


Figure 1 'Spot' sample dates and locations in the Frome floodplain. Temporal data sets were collected at site A and at Ds. Three categories of pool were recognised - 'Connected,' -those which had received river water prior to sampling but were separated from the river when the sample was taken (green); 'Flowing', sites receiving river water at the time of sampling (blue) and 'Isolated', pools which were never connected to the river except in very extreme flood conditions (red). Photos depict upstream sample areas in March 2014 showing re-deposited sand near the flowing pool at L9 and stagnant pools near S1.

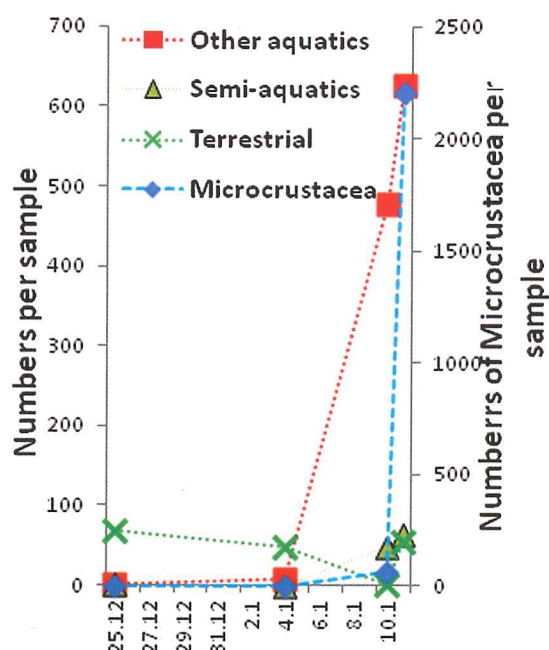


Figure 2 Fluctuations in the numbers of macroinvertebrates at Site A.

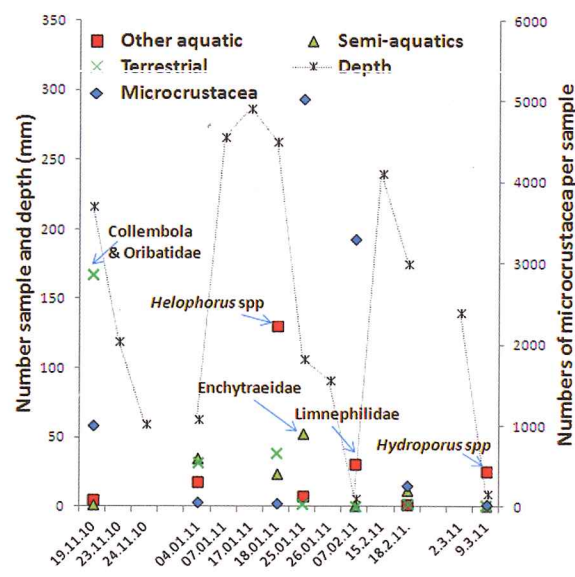


Figure 3 Fluctuations in macroinvertebrate numbers at Site D (L1 and S5) and water depth at time of sampling.

were the most abundant aquatic component. In February the trichopteran species *Limnephilus auricula* and *L. griseus* had replaced Coleoptera as the dominant aquatic taxa to be succeeded by the larvae of the coleopteran *Hydroporus* in early March just before the habitat dried out. The semi-aquatic enchytraeid worms were present throughout January and into February.

Spot Samples

This data set comprised six isolated pools. Four of these were sampled on more than one occasion and included Pool D from which samples L1 and S5 were taken (Figure 1). The fifth pool, K1, was connected to the river only at extreme flood levels. Site C, (4 km upstream) was a flooded meadow unconnected to the main river.

The remaining sites were small patches of water originating from a combination of heavy rainfall and river bank overtopping, including 'connected' pools which had received river water prior to sampling, but were separated from the river when the sample was taken, as well as 'flowing' pools receiving river water at the time of sampling.

Variation in the collecting method and operator between samples collected in January and February in 1999/2000 and the remaining samples preclude the use of sophisticated statistical analyses. We simply present faunal composition data in relation to pool type and season.

Although there are some trends in the data set (Figure 4), such as higher abundance of microcrustacea and coleopterans in isolated pools, greater numbers of gastropods and crustaceans in connected pools and a generally higher diversity in the flowing habitats, there is considerable variation even within pool types.

This variation is attributable to season and location. In isolated pools, Diptera (Culicidae and Chironomidae) were common in December samples and microcrustacea dominated sites in March.

Diptera (Simuliidae) again dominated the flowing sites in December but Ephemeroptera (*Baetis* spp) were abundant in March. In connected pools, November samples were characterised by high numbers of gastropods including *Aplexa hypnorum* and *Galba truncatula*.

The December samples taken further upstream had few gastropods and the March series had two sites in which gastropods were common - mainly *G. truncatula* but with the addition of taxa such as *Bythinia*, *Physa* and *Theodoxus fluviatilis* more often found in running water.

The crustacean *Crangonyx pseudogracilis* was more common in November and December than *Gammarus pulex* which occurred more frequently in the March samples.

C. pseudogracilis is a North American species first recorded in the UK in 1935. Despite having no strategies for survival in temporary waters (resting eggs, dormant juvenile stages or active burrowing), it has ecological traits such as underground dispersal abilities, an annual life cycle, tolerance to low amounts of oxygen and the ability to survive for up to five months in soil with an average moisture content of 51 % (DiSalvo and Haynes, 2015). These characteristics have probably aided its distribution throughout the UK and its success in temporary pools.

All pool types were physically similar, grassy-bottomed shallow water bodies, but the fauna that developed within them was dependent on a number of variables including:

- The topography of the terrain and exposure
- Proximity to the river
- Contribution of river water



Plate 1 Fluctuations in water level at Pool D (L1 and S5) between 17 January and March 9 2011.

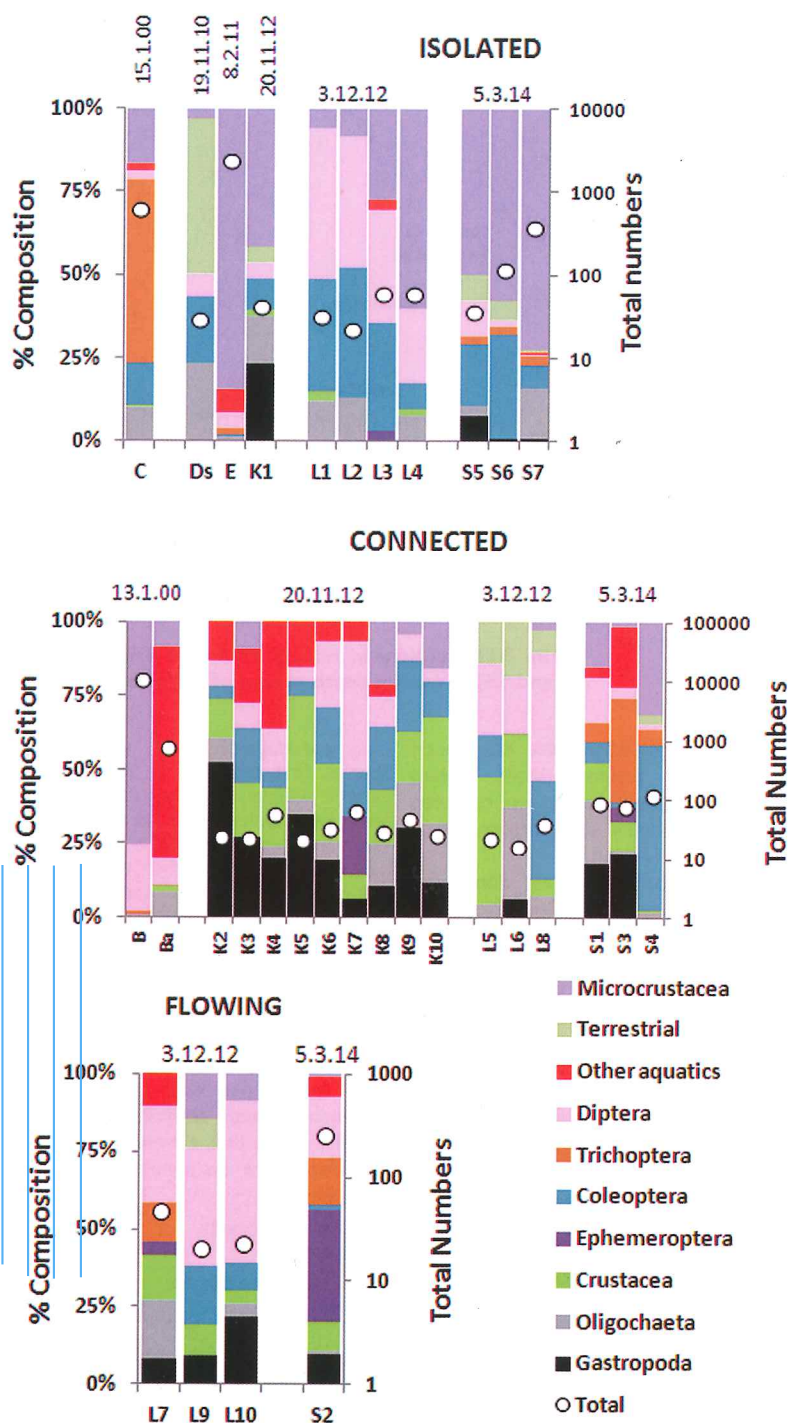


Figure 4 The faunal composition of 'spot' samples taken on the River Frome floodplain in three categories of pool type. Total numbers are presented on a log scale

- Time since flooding – how long has the habitat been wet
- The season of flooding
- Climatic conditions at the time of the flooding; and subsequently
- The year of sampling

All these factors interact and result in a series of pools of differing age which have experienced a range of climatic conditions and which are available for colonisation

by whatever organisms are active. We might have expected to find a core pioneer community and this might have been the case if all the pools were formed at the same time and sampled from first flooding but it is clear that the factors listed above work together to influence faunal community composition and the ephemerality of the habitat will control the stage of development of the macroinvertebrate community.

The two sets of temporal observations indicate clearly the truncation of faunal

development following drying. The extreme conditions do not favour the development of a rich fauna for individual pools (8-20 taxa in Nov. to Jan. and 16-34 in March) but overall, 153 taxa were recorded. A recent study (Armitage *et al.*, 2012) of tyre tracks and puddles has shown that these unprepossessing habitats can together support high biodiversity (174 taxa) and the flooded pools examined here have shown a similarly high value.

Management regimes which aim to reduce topographic variability (levelling land and infilling depressions) will reduce the mosaic of small temporary habitats which facilitate the spread and survival of freshwater taxa and maintain the diversity of the floodplain.

Although the literature on temporary waters is extensive (Williams, 2005) these ephemeral habitats in the Frome floodplain offer an opportunity for a more detailed examination of the mechanisms of colonisation and faunal development. At present, we simply have a description of the fauna of pools at various stages of development which, as is often the case, raises more questions than it answers.

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